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Forecasting multi-frequency intraday exchange rates using deep learning models

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Abstract

This paper examines the behavior of currencies' intraday exchange rates with mainly focuses on predicting these behaviors through deep learning models. The time series data are used in this study and comprise intraday exchange rate data for seven volatile currencies, recorded at two different frequency intervals: 1 h and 30 min. The data cover the time frame from January 1, 2018, to December 31, 2020. Firstly, wavelet maximal overlap discrete wavelet transform (MODWT) type "haar" is applied in order to identify the noise representing the volatile trend. Then learning models are applied to historical data that includes support vector regression (SVR), recurrent neural network (RNN), and long short-term memory (LSTM). The main findings of the study provide strong evidence that the deep learning technique (i.e., LSTM) outperforms the other compatible models. This has been confirmed by the statistical measures for accuracy purposes. This study intends to address complex trends exhibited by volatile behavior of intraday exchange rates through non-conventional learning techniques for the time of need-based for the currency traders and other stakeholders.

KEYWORDS

forecasting, intraday exchange rates, LSTM, MODWT, multi-frequency

1 | INTRODUCTION

Forex markets are the intuitive part of global financial markets with a daily trading volume of \$7.5 trillion (Wang et al., 2023). The evolution of these financial markets makes them chaotic, volatile, and complex nonlinear in nature. These characteristics in addition to high-frequency data make conventional forecasting techniques sometimes unreliable in terms of favorable forecasting results (Deng et al., 2022; Li & Bastos, 2020). Traditional methods such as linear regression and autoregressive integrated moving average (ARIMA) models may not be sufficient to capture the intricate dynamics of the market. Additionally, the increased interconnectedness of global

financial systems means that unexpected events in one market can have ripple effects on others, leading to even greater unpredictability. Under this environment, reliability and accuracy of forecasting models is difficult and under a big question mark (Cao et al., 2019). Studies like Hsu et al. (2016), Kewat et al. (2017), Khairalla and AL-Jallad (2017) show that nonconventional techniques outperform the conventional techniques of forecasting financial markets.

Recently, different kinds of studies analyzed the presence of chaos in the exchange rate as well as other financial or economic time series, and this chaos is tested by different means and objectives accordingly. Nicolis (1998) explains complexity as an ability to switch between

models of behavior and depends on environmental conditions. Complexity in exchange rates and their smoothness require the feature, model selection, and parameter tuning and provide a significant impact on prediction tasks. Therefore, each step in the process requires successful optimization and overcoming significant complexity, and it is not easy without complexity knowledge and proper analysis (Yao et al., 2017). Since last 15 years “Computational intelligence” has gained a lot of popularity as a technique for exchange rate prediction. Artificial neural networks (ANNs) are specially used for the purpose as well as in different fields like business and industry (Abiodun et al., 2018; Bellgard & Goldschmidt, 1999; Cohen, 2022; Kumru & Kumru, 2014). In computational intelligence, the more common technique is that it uses generalization by nonlinear approximation and in high dimensional spaces usually (Kecman, 2001). There is a powerful generalization that produces output by using input those are unseen, which is done by previously captured patterns and learned inputs, due to this ability these techniques are outperforming regression models (Kecman, 2001).

Many studies have verified that the financial time series data are time-varying along with characteristics of nonlinearity and stochastic volatility (Alfarano et al., 2008; Luo et al., 2018; So et al., 2002). As stated by Hull (2012), volatility is the standard deviation of a variable's return that is provided by the per unit of time using continuous compounding. Risk is managed by the use of volatility, and the unit of time is 1 day (Hull, 2012). There is contention among financial economists, and the reason behind this is the highest accuracy of the volatility estimation (Peng et al., 2018). It is observed by Nelson (1992) and Merton (1980) that there is no need for huge historical data for the forecasting of volatility; instead, it is enough to have a short observation period for the volatility forecast (Andersen et al., 2003). There is progress in the use of intraday data for volatility forecasting because accurate predictions can be made with the help of data that are of a short time (Peng et al., 2018). The increase in volatility of the exchange rate affects the economy negatively as a result of which the cost of international transactions is increased (Baum and Caglayan, 2010), the business profitability is also affected (Braun & Larrain, 2005), and domestic as well as foreign investment risk also gets increased (Urata & Kawai, 2000).

Forecasting the foreign exchange rate is an ongoing challenge for professionals, policymakers, and academics. There are large numbers of studies that are dedicated to the dynamics of the exchange rate, and from those studies, two main conclusions can be made: one is that macroeconomic fundamentals should be dictated in the theory about the purchasing power of a country's

currency because the macroeconomic fundamentals are less useful for the prediction of the exchange rates (Bacchetta & Van Wincoop, 2004).

Secondly, a proper modeling technique or a proper set of information can be helpful in the short-term forecasting of the exchange rate movements like order-flow of customers can be taken as an example; it can be used as information (Bacchetta & Van Wincoop, 2004). The direct application of the advanced time series models has some difficulties though they are useful in out-of-the-sample forecasting of the density; for the trading strategy of traders and the quotation strategy of dealers, a successful practice model for the forecasting of the mean is much more useful that it can be directly applied to them (Tsang et al., 2007).

The adoption of newly developed machine learning (ML) and artificial intelligence-based forecasting techniques counters these arguments with more accurate forecasting results (Kumbure et al., 2022). There is a plentiful and vast literature on the prediction of exchange rate volatility (Zhou et al., 2020). The motivation of the current study is since, for the prediction of the financial market, ML is a new technique and widely used as advised by researchers (Kara et al., 2011; van Gerven & Bohte, 2017; Q. Zhang & Yan, 2018). And researchers still try to figure out the best prediction method among machine and deep learning methods. Although, studies, that is, Jiao et al. (2020) and Zhao et al. (2019), claim the effectiveness of ML and deep learning techniques for forecasting financial markets as these techniques boost the prediction performance.

Peng et al. (2018) used support vector machine (SVM) for the forecasting of volatility. Support vector regression (SVR) aims to find out the function of decision that is best for approximation of observation. Historically, developed soft computing methods, that is, SVM by Vapnik (1995), are used to separate and label the data and are known for their simplicity. By using the structural risk-minimizing principle, SVM minimizes the upper bound of generalization error. Many studies adopted SVM techniques to predict the financial time series and have found better predictors based on its properties (Alhnaity & Abbod, 2020; K.-j. Kim, 2003; Tay & Cao, 2002). Exchange rates in forex markets are a reflection of all current and previous information, and with simple neural networks, the earlier information may be lost. Recurrent neural network (RNN) often faces the vanishing gradient problem while using certain activation functions. To deal with this vanishing gradient problem, Schmidhuber and Hochreiter (1997) proposed a long short-term memory (LSTM) network. A sophisticated LSTM network gives some astonishing results in sequence learning tasks. LSTM network extends the

learning memory that helps the network to remember the important pattern of information in the network. This network can read, write, and delete the information that is stored in memory. Yan and Ouyang's (2018) study found that LSTM is a better method to predict the daily returns of the Shanghai composite index, and another study by Hu et al. (2020) used LSTM to forecast volatility and argued that ML is a process to boost the performance of prediction. These ML models not only have the ability to work independently, but also, many of the studies used these ML models to modify or to create hybrid models with classic forecasting techniques.

Researchers, different firms, banks, and industries use machine and deep learning techniques for their prediction and forecasting purposes so there is a need to find out the best prediction model among machine and deep learning models being used. There are many folds of the irregularities found in the data, and these irregularities change with time by using the different scales (Fan & Wang, 2007). To find out irregularities wavelet method is used it tells us about the real trend in the original data.

Wavelet is an ideal tool for the studying of the non-stationary time series because it adapts itself very intelligently for capturing the features that are around the wide range of the frequencies and the local time events are captured (Afshan et al., 2018). "Wavelet," as its name suggests, is a small wave. There are different types of wavelets like CWT, DWT, and maximal overlap discrete wavelet transform (MODWT) (Childers, 1978). CWT and DWT are the older versions and do not identify the signal with all and proper information because they are not detailed versions of the mother wavelet for decomposing the original series (Gençay et al., 2003). The series in DWT wavelet may not be accurately aligned. To overcome these problems in wavelet, MODWT has been proposed (Deo & King, 2015). This study uses MODWT for trend identification in the currencies used because this wavelet transform can handle any type of data and it has increased resolution as compared with other types like DWT. This study used MODWT analysis for trend identification by using Haar function of wavelet because Haar provides the proper and smooth wave in the time series.

Forecasting is a never-ending debate. There is growing evidence to suggest that deep learning models can be highly effective for forecasting financial time series data, including multi-frequency intraday exchange rates. One advantage of ML techniques is their ability to capture complex patterns and relationships in the data that may not be easily identified using conventional techniques. This can be particularly useful in financial markets, where there are often many factors at play that can influence exchange rates, including macroeconomic

variables, geopolitical events, and market sentiment (Khan et al., 2023). Furthermore, these ML techniques can be trained on large amounts of data and can learn from these data over time, making them well-suited to forecasting tasks where the underlying patterns may change over time. This can be particularly important in the forex market where exchange rate dynamics can be highly dynamic and subject to sudden shifts. Historically, we have witnessed several studies conducted to forecast the exchange rates using both conventional and unconventional forecasting techniques (Al-Gounmeein & Ismail, 2020; Alhnaity & Abbod, 2020; Frankel & Froot, 1990; Huang et al., 2010; Ince & Trafalis, 2006; Kamruzzaman & Sarker, 2003; Sun et al., 2020; Tenti, 1996). However, a comparative study across the market for the intraday multi-frequency time series is missing in the literature. This study is aimed to fill the gap that existed in the work of Yao et al. (2017) through bridge the gap in relation to forecasted model performance based on non-conventional time series model techniques and in the following ways.

The use of non-conventional forecasting techniques based on machine and deep learning has been gaining popularity in recent years due to their ability to capture complex patterns in data that traditional statistical methods may miss (Zhang et al., 2017; Zhang & Yan, 2018). Firstly, this study contributes toward introducing the non-conventional models of time series based on machine and deep learning models by using two different frequency intervals of intraday, which have been adopted by previous studies, that is, Aloud (2020) and Li et al. (2020) to forecast stock markets prices.

Secondly, the study tested for the noise through the application of MODWT and denoised it for further analysis. For comparison purposes, the accurate measurement of these forecasting techniques is an important part, and studies, that is, Erçen et al. (2022), Flores (1986), Li et al. (2020) and Sun et al. (2020), applied these statistical measures to compare the forecasting techniques. So thirdly, to evaluate, compare, and assess the resulting outcomes from these models, accuracy indicators of statistical measures have been used for identifying the best prediction model.

This paper has the contribution toward practical implications for research scholars for using another range of machine and deep learning techniques to predict the volatile currency market movements of similar nature. Fourth, the study proposed to advise investors to take better investment decisions in order to diversify risk in the currency market through model selection and evaluation criteria that contributes further to understanding the complex nature of currency market forces for entangling all time dynamics of forecasting techniques.

The findings of the study suggested that to forecast the intraday exchange rates, LSTM was found to be the best forecasting technique out SVM and RNN based on its predicting power to handle large-scale data and extract hidden information. The current study finds are align with previous studies by Abedin et al. (2021), Escudero et al. (2021), Ranjit et al. (2018), Sezer et al. (2020), Sun et al. (2020), and Yan and Ouyang (2018). The findings of the study will be helpful for banks and currency traders in the financial industry. They will be able to use the best prediction technique that will be more accurate with less error among machine and deep learning techniques for the multi-frequency time series in currency markets.

The study is organized in the following manner—Section 2 represents the data selection and descriptive analysis; Section 3 discusses the methodology based on selective non-conventional time series models; Section 4 represents the analysis and robustness of the study; Section 5 discusses the practical implications before concluding the paper.

2 | EXPLORATORY DATA ANALYSIS

This study used two different frequency intervals of intraday, that is, data of time series for 7 currencies against the dollar, that is, exchange rate based on the availability of data. The targeted currencies for this study are gathered from [dukascopy.com](https://www.dukascopy.com) because it provides complete and clean data sets. Currency exchange rate data are 3 years ranging from the period January 1, 2018, to December 31, 2020, and the selection of timeframe is based on the limitation of data as the aforementioned source does not allow for retrieval of data more than 3 years. Also, studies like Li et al. (2020) used 2 years of data to train and test the deep learning models. The time series data selected for the study are intraday currency exchange rates with a frequency interval of 1 h and 30 min.

2.1 | Data pre-processing

The primary part of data analysis is to clean the data and make the data fit for training. There are different steps to be taken in data preprocessing, and data must be looked at before processing it in any statistical model (Granger, 1981). Following are the preprocessing actions that have been conducted in this study. Before computing the ML techniques, it is mandatory to scale data because if the data are not scaled, it create problems for model fitness. After all, the large ranges in data learning speed are

TABLE 1 List of currencies with description.

Currencies	Description
USD/CAD	US dollar vs. Canadian dollar
USD/CNH	Us dollar vs. Offshore Chinese Renminbi
USD/CZK	Us dollar vs. Czech Koruna
USD/HUF	Us dollar vs. Hungarian Forint
USD/JPY	Us dollar vs. Japanese Yen
USD/MXN	Us dollar vs. Mexican Peso
USD/NOK	Us dollar vs. Norwegian Krone

reduced, and the accuracy of the model also gets effect. So, the data are scaled by using the “min-max scaler” object that is from scikit-learn library. It transforms our data in the range of 0–1. This object takes the minimum value of the column as 0 and the maximum as 1, and it fits all other values on its basis.

Table 1 represents initial data exploration based on central tendency measures along with the normality test statistics a.k.a. Jarque-Bera (J-B) test statistics. For data location measurement, we simply interpret the skewness and kurtosis and evaluate the concentration of observed values. In this data set CNH, HUF, and JPY have shown negative trends, and CAD, CZK, MXN, and NOK have shown positive trends in values based on skewness.

2.2 | Wavelet-MODWT

In this step, MODWT is applied to each currency in a 1-h data set. This study chooses MODWT because it is a more effective wavelet version as discussed above that can handle data of any length and it captures more information in each scale. Following are the figures (graphs) of each currency original series along with the MODWT series. In order to investigate the trend of all 7 currencies' prices, this study conduct wavelet MODWT analysis to address the trend in each currency price. Mentioned are the graphs of the original series and series after applying wavelet MODWT analysis for the selected currencies, and every figure shows the original data series of a currency and its wavelet-analyzed series of the currency against it. It is clear from the MODWT analysis of every currency that each currency contained a specific trend (see Figure 1a–g).

Therefore, it is clear from the wavelet analysis that CAD, CNH and CZK were having a downtrend in their price in the year 2018, and further in 2020, they reached a maximum point in their trend. HUF, MXN and NOK were observed to have fallen in their prices in 2019 as shown in the trend, and they reached to peak in 2020.

TABLE 2 Descriptive statistics.

Currencies	Mean	Median	Max.	Min.	SD	Skew.	Kurt.	J-B
CAD/USD	1.3212	1.32	1.466	1.2259	0.0348	0.5325	4.5737	3957.374**
CNH/USD	6.8109	6.8779	7.1945	6.2408	0.2483	-0.7521	2.5153	2737.106**
CZK/USD	22.6176	22.6975	26.0628	20.1071	1.0431	0.1766	3.9619	1150.799**
HUF/USD	289.536	289.674	340.634	246.805	18.8009	-0.2126	2.8583	220.185**
JPY/USD	108.72	108.671	114.532	102.067	2.5035	-0.0366	2.3323	494.4206**
MXN/USD	19.9832	19.3553	25.631	17.9454	1.5271	1.437	4.258	10786.84**
NOK/USD	8.7831	8.7098	12.0538	7.6356	0.6597	0.9318	4.6825	6909.425**

Note: Table 1 contents include currencies exchange rates (see Table 1 for detailed description) and measure of central tendency and measure of normality, that is, J-B = Jarque-Bera test statistics.

***Confidence interval of p -value at 1% level.

**Confidence interval of p -value at 5% level.

For CHF and JPY, the trend shows that their prices were at a peak in 2019, and they started falling after it and reached a minimum in 2020 as shown in their respective trends. The wavelet analysis shows that there are a lot of fluctuations in the prices of currencies, and this study aims to investigate the future price of currencies by using highly effective machine and deep learning model techniques to predict the prices and to address the predictability power of ML and deep learning models that which model among machine and deep learning performs well, by using selected accuracy indicators statistical based measures.

2.3 | SVR

According to statistical learning theory, SVR is one of the popular supervised learning techniques for nonlinear regression analysis. According to the artificial neural network, SVR is a (ML technique) non-conventional model that is represented as x proposed by Vapnik (1995).

$$f(x) = (Z \cdot \phi(x)) + b, \quad (1)$$

where $\phi(x)$ is kernel function that is used for the transformation of nonlinear input data into linear data. Z represents the weight vector, and b represents the bias. SVR method aims to find out a decision function for the observation sets according to the best approximation (Müller et al., 2005; Vapnik, 1995). SVR method is used for solving the problem of pattern classification. Whereas, regression functions for SVR in feature space are represented as

$$f(x; w, b) = Wt\phi(x) + b, \quad (2)$$

where ϕ is the input feature x , b represents the threshold, and the weight set in the above equation is denoted by w .

2.4 | RNN

RNNs have shown important capabilities of prediction of sequence in recent studies RNN retain information from past data for using the information of data to predict future values. RNNs can use an internal process state as the input sequence. Among all NN models, RNN known as RNNs makes the time series analysis more significant and adaptable (Galeshchuk, 2016; Zeng & Khushi, 2020).

$$mn = f(Wxn + b, -1 + bm), \quad (3)$$

$$yn = f(Vmn + by), \quad (4)$$

where f is the function for activation and W , b , V , and b are the parameters, matrices, and vectors. For financial time series sequence, correlation is an important feature used, and this is indispensable when the model for the prediction is established.

2.5 | LSTM

Deep learning neural networks have a type called LSTM; they are the structure of RNNs improved. In standard RNNs, LSTM is used for the solution of long-term dependence as well as gradient disappearance problems caused in RNNs; this is done by a concept of memory units that include completely trained

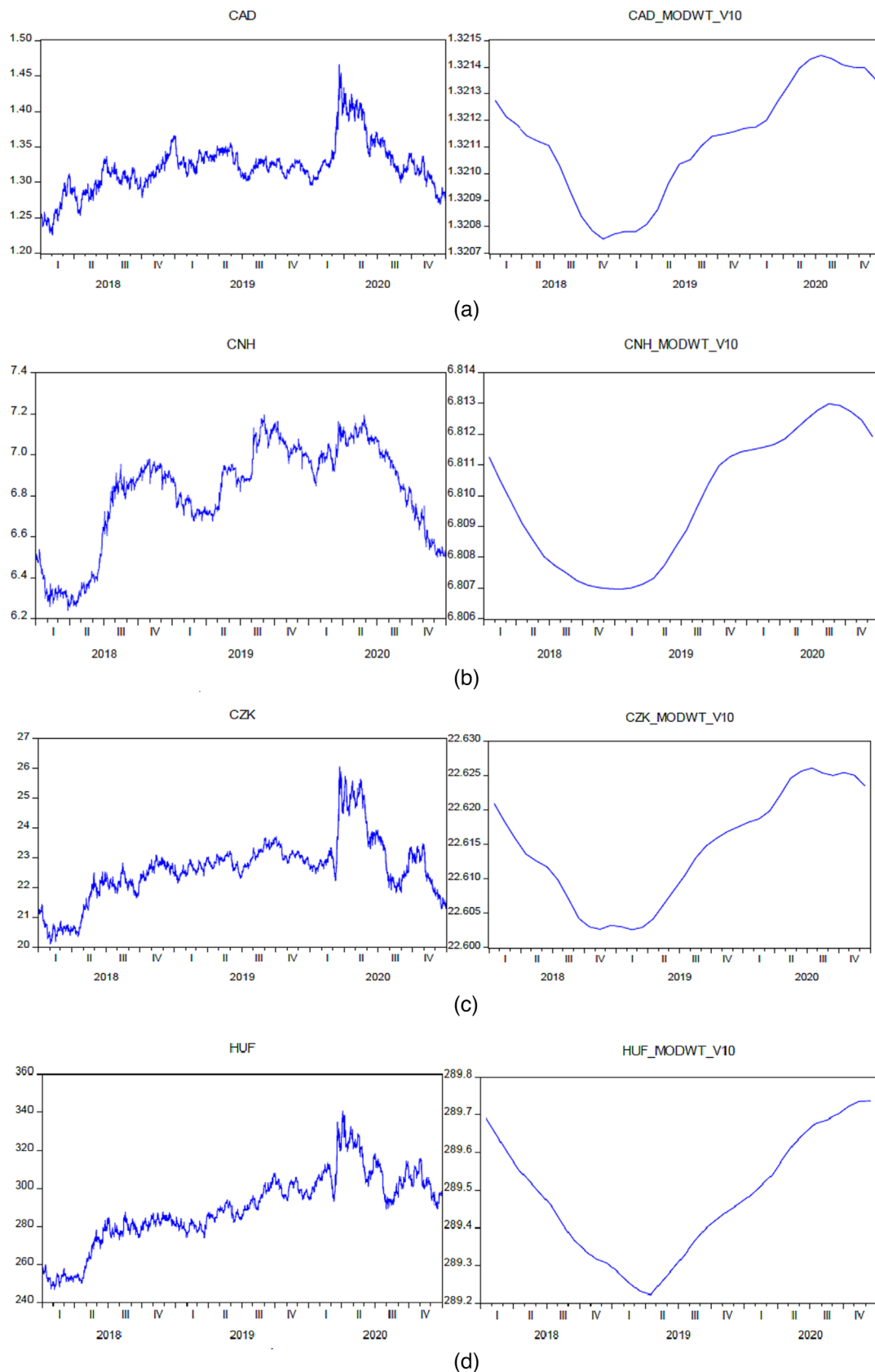


FIGURE 1 (a) CAD and original series versus maximal overlap discrete wavelet transform (MODWT). (b) CNH original series versus MODWT. (c) CZK original series versus MODWT. (d) HUF original series versus MODWT. (e) JPY original series versus MODWT. (f) MXN original series versus MODWT. (g) NOK original series versus MODWT.

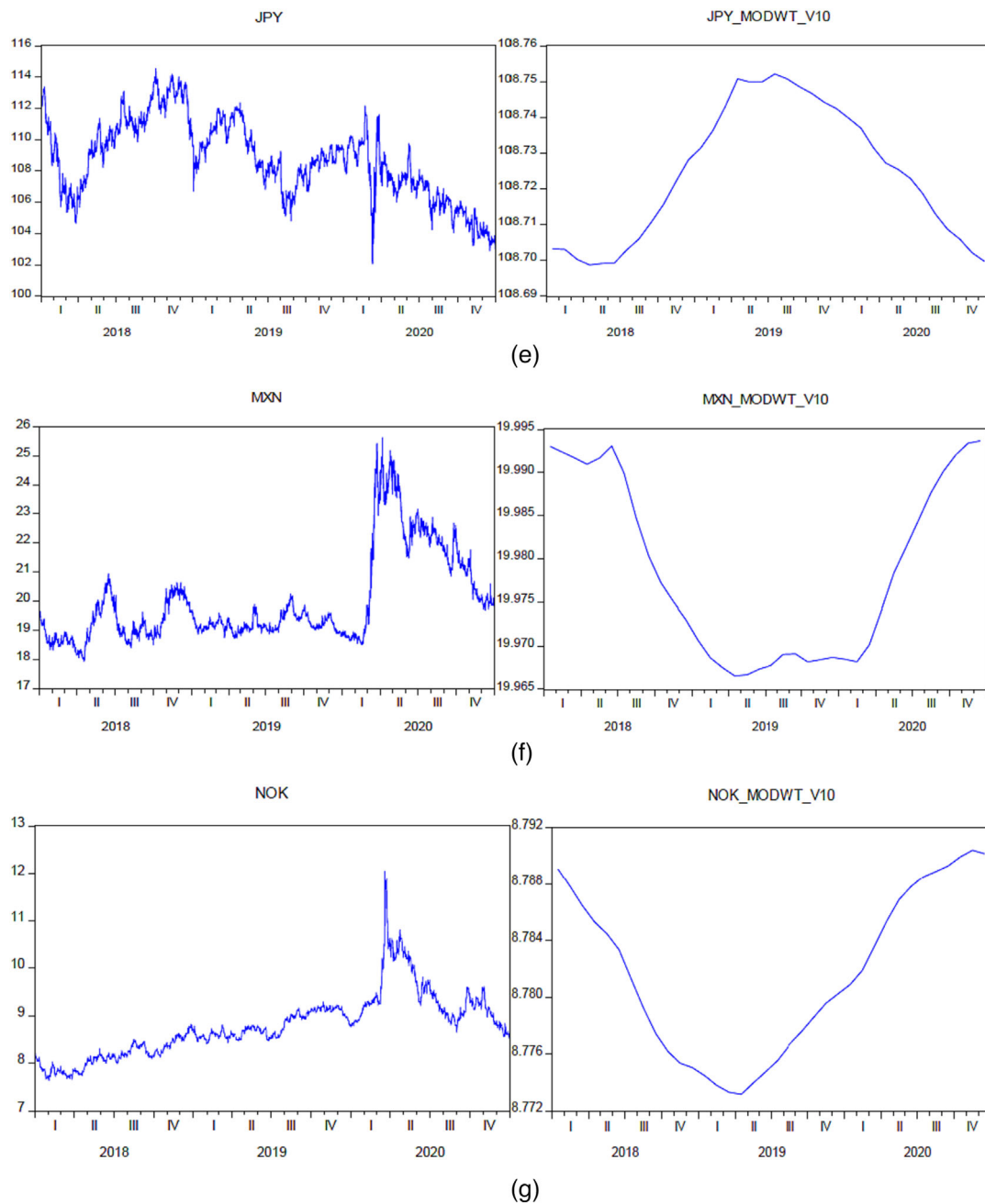


FIGURE 1 (Continued)

models and that model has a satisfactory effect in the process of time series analysis (Kim & Won, 2018). For time series data, LSTM is the most suggested method. This study will use LSTM for the analysis of multi-frequencies of time series for checking the accuracy of the results in the selected data set of time series.

$$it = \sigma(w[h(t-1), xt], bi). \quad (5)$$

The modulation gate of the input is represented as

$$ct = \tanh(w[h(t-1), xt], bc), \quad (6)$$

where in the above-given equations, weight and input of the values are represented by it and ct , respectively. The other added parameters are the same as the previous. Though LSTM input gate consists of both input and modulation gate, it is the combination of these two. Three-output gate in the model of LSTM is used for the decision of hidden layer that which of the portion from the current cell will go to the next hidden layer. The output gate equation is represented as follows:

$$ot = \sigma(w[h(t-1), xt], bo). \quad (7)$$

And the equation for the hidden state is as under:

$$ht = ot * (ct). \quad (8)$$

In the output gate, the activation function is used for the reason of value selection between 1 and -1 . Like the input gate, output gate also consists of two functions, and it is a combination.

3 | MODEL ANALYSIS

This section of data analysis deals with ML and deep learning techniques. This section's techniques need first of all to set an environment and then proceed to data pre-processing after that data are processed through ML and deep learning models.

3.1 | Setting environment and parameters

This study used some tools for experiments and to develop some prediction models for analysis purposes. This study chooses Python as a programming language to set environments, and Jupyter Notebook is used as IDE for Python. Jupyter Notebook is an application that is used for code creation, and different programming languages can be supported by Jupyter Notebook. Libraries used are Numpy for the mathematical functions because the prediction models need Numpy arrays as parameters for fast operations and to decrease the time of training and prediction. Pandas are used for the manipulation of data and analysis as they provide data structure and analysis tools for operations. Matplotlib is used as a plotting library for the creation of graphs. scikit-learn is used for stat models as scikit-learn is focused on ML and it provides us with scaling the data into train and tests portions and as well as it is used for mean square error (MSE) as evaluation. Keras and Tensor Flow library are used for ML operations as they can do the task of normalization and classification. Keras is used for deep learning operations.

In the initial part, three distinct models, that is, SVR, RNNs and LSTM techniques, representing supervised and unsupervised learning models have been applied, respectively. For ML, data set is divided into two parts that are the training and testing portion, and their ratios are set to 70:30, which means that 70% of the data is used for training and 30% portion of the data is used for testing. For ML, this study uses SVM and as it is regression-

based forecasting so Kernel function is an essential parameter in SVR. In the scikit-learn library, there is a range of kernel functions, but RBF is a common kernel function because it can deal with small and larger data samples, as well as it can map the data sample into high dimensions and have the characteristic of low parameter requirement. On the bases of said characteristics (Kim, 2003; Li et al., 2020; Min & Lee, 2005), literature supports RBF as kernel function for ML SVR. The other parameters like c , gamma, and epsilon are set as default. This study takes the time variable as input for support regression analysis.

ML technique works on pre-defined rules and data features, but deep learning technique can identify data features from raw data. In deep learning, LSTM and RNN neural layers are important; therefore, two layers of neurons are used in this study. The first layer consists of 64 neurons, and dense neurons are 20, whereas the second layer consists of 32 neurons and 1 dense neuron for output generation. An activation function is needed for the optimization of neural networks, the output of the neural network, efficiency and accuracy depends on the activation function used. Due to more properly tuned results, tanh activation function outperforms other activation functions (Li et al., 2020); "tanh" is used as an activation function that is similar to the logistic sigmoid, and it ranges from -1 to 1 , which made it better. RMSprop is used as an optimizer function because it increases the learning rate and it speeds up the loss computation process, and MSE is used to define the loss.

The batch size has been set to 256 with a buffer size set according to the data. The evaluation interval in every epoch is set as 250, and epoch's number is set as 100. Figure 2a-g represents the prediction visualization for SVR, Figure 3a-g for RNN, and Figure 4a-g prediction graphs for LSTM approach for exchange rate time series of 1-h frequency interval. Tables 1-4 summarize the statistical analysis of performance metrics of selective non-conventional models with given accuracy indicators such as MSE, root mean square error (RMSE), mean absolute error (MAE), and mean absolute percentage error (MAPE). Studies like Erçen et al. (2022), Flores (1986), Li et al. (2020), and Sun et al. (2020) used these evaluation criteria to measure the accuracy of forecasting techniques. For a better understanding, the statistics are given in these tables shown in bold.

Figures 2a-g to 4a-g show the forecasting accuracy on the timeline bar, and Tables 3-6 draw the following conclusions. First, from the three benchmark models, an unsupervised model, that is, LSTM, fits in a nonlinear manner for respective currencies exchange rates, with no inferior performance of RNN for all cases. Because, between all NN models, RNN makes the time series

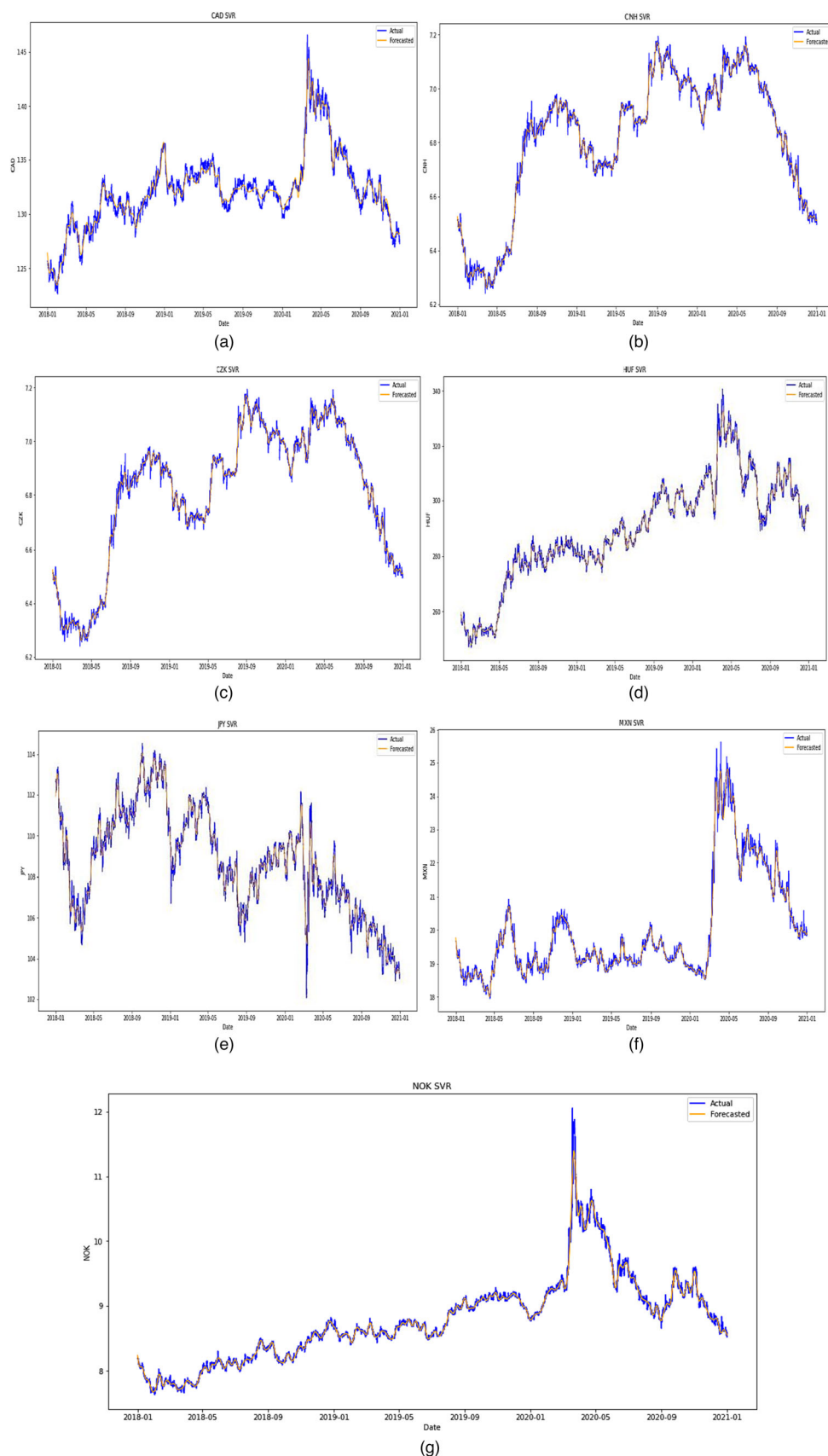
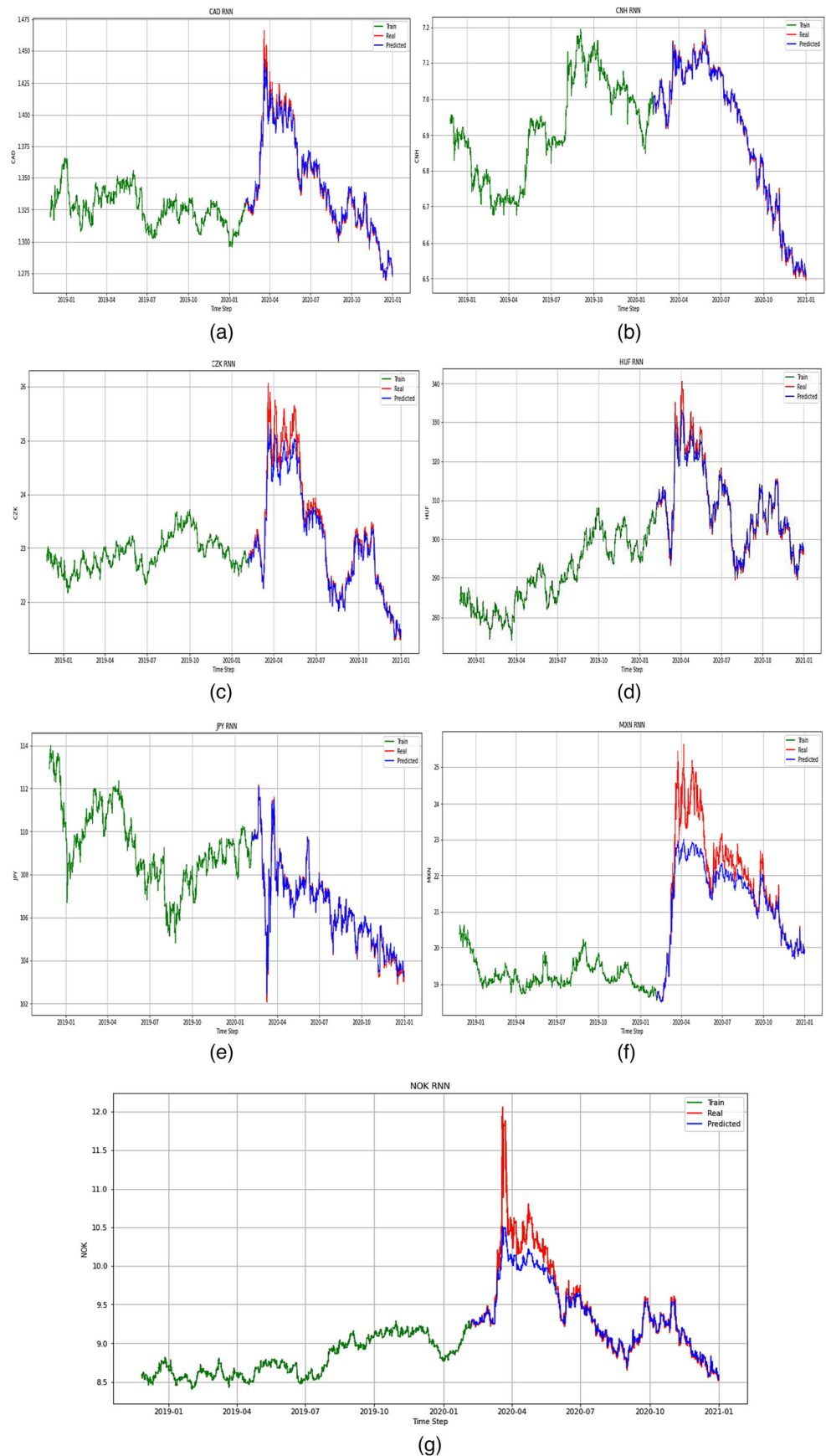


FIGURE 2 (a,b) CAD and CNH actual and predicted prices with support vector regression (SVR). (c,d) CZK and HUF actual and predicted prices with SVR. (e,f) JPY and MXN actual and predicted prices with SVR. (g) NOK actual and predicted price visualization with SVR.

analysis more significant and adaptable (Zeng & Khushi, 2020). Moreover, RNNs have shown important capabilities of prediction of sequence in recent studies,

that is, making prediction task more efficient (Qin et al., 2017). Second, the LSTM managed to obtain the lowest error score for the given accuracy indicators, MSE,

FIGURE 3 (a,b) CAD and CNH actual and predicted prices with recurrent neural network (RNN). (c,d) CZK and HUF actual and predicted prices with RNN. (e,f) JPY and MXN actual and predicted prices with RNN. (g) NOK actual and predicted prices with RNN.



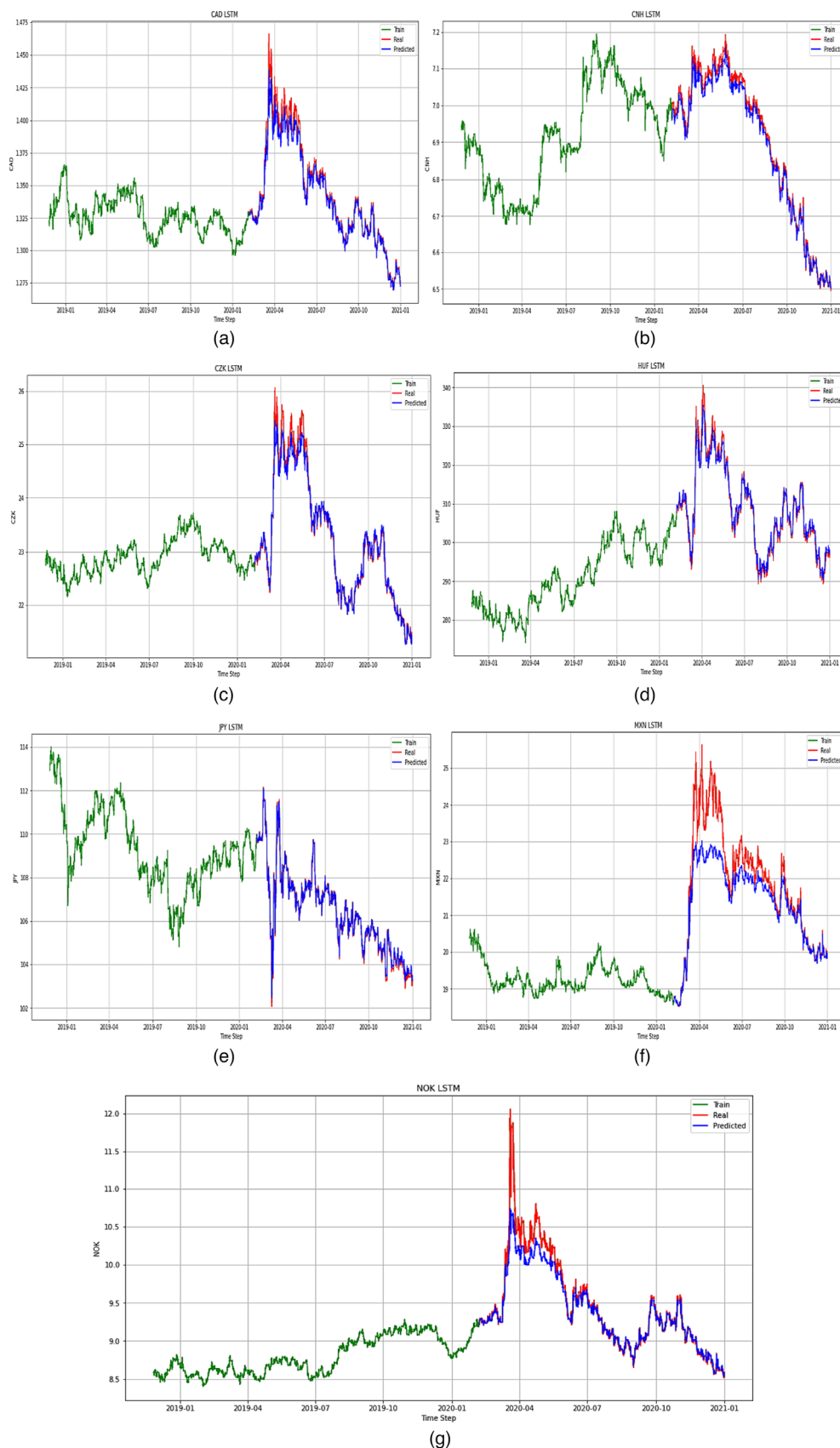


FIGURE 4 (a,b) CAD and CNH actual and predicted prices with long short-term memory (LSTM). (c,d) CZK and HUF actual and predicted prices with LSTM. (e,f) JPY and MXN actual and predicted prices with LSTM. (g) NOK actual and predicted price visualization with LSTM.

RMSE, MAE, and MAPE. LSTM error score ranges between $[0, -2]$ for given behavior of exchange rates nonlinear trends. In particular, when judging from the

MSE accuracy indicator, the lowest error score of prediction is for Yuan, Czech koruna, and Yen, respectively. Third, the statistics results from the four accuracy

TABLE 3 Performance comparison of models based on MSE test statistics.

Inputs	SVR	RNN	LSTM	Best model
CAD/USD	0.0012	1.1145	1.4111	SVR
CNH/USD	0.0619	3.6575	0.0002	LSTM
CZK/USD	0.0619	0.0448	0.0172	LSTM
HUF/USD	348.858	1.969	1.2479	LSTM
JPY/USD	6.2519	0.0144	0.0135	LSTM
MXN/USD	2.2985	0.5065	0.5065	RNN/LSTM
NOK/USD	0.4257	0.0474	0.031	LSTM

Abbreviations: LSTM, long short-term memory; MSE, mean square error; RNN, recurrent neural network; SVR, support vector regression.

TABLE 4 Performance comparison of models based on RMSE test statistics.

Inputs	SVR	RNN	LSTM	Best model
CAD/USD	0.0346	0.0033	0.0058	RNN
CNH/USD	0.2487	0.006	0.0167	RNN
CZK/USD	0.2487	0.2116	0.1312	LSTM
HUF/USD	18.6777	1.4032	1.1171	LSTM
JPY/USD	2.5004	0.12	0.1162	LSTM
MXN/USD	1.5161	0.7117	0.7117	RNN/LSTM
NOK/USD	0.6524	0.2176	0.176	LSTM

Abbreviations: LSTM, long short-term memory; RMSE, root mean square error; RNN, recurrent neural network; SVR, support vector regression.

indicators further confirm the strength of the LSTM technique in predicting the high volatility trends shown in the observed time series. This predicting ability of LSTM compared with other benchmark models is superior in extracting hidden information for these intraday exchange rates of selective currencies. This again supports the argument developed in literature by Abedin et al. (2021), Escudero et al. (2021), Ranjit et al. (2018), Sezer et al. (2020), and Sun et al. (2020).

3.2 | Robustness test

In order to ensure the reliability and accuracy of the model, additional robustness test is necessary. Forex markets are dynamic and chaotic, which makes forecasting a complex and challenging task, and also there is a multitude of factors that can influence exchange rates. Investors these days hold access to high-frequency data. This high frequency holds more market information (Andersen et al., 2003). This robustness check helps to determine the reliability of the model at more high

TABLE 5 Performance comparison of models based on MAE test statistics.

Inputs	SVR	RNN	LSTM	Best model
CAD/USD	0.0245	0.0023	0.0036	RNN
CNH/USD	0.2039	0.0046	0.0141	RNN
CZK/USD	0.2039	0.1358	0.0768	LSTM
HUF/USD	14.8264	0.9482	0.8264	LSTM
JPY/USD	2.0597	0.0822	0.0744	LSTM
MXN/USD	1.1722	0.46	0.4598	RNN/LSTM
NOK/USD	0.4982	0.1025	0.0808	LSTM

Abbreviations: LSTM, long short-term memory; MAE, mean absolute error; RNN, recurrent neural network; SVR, support vector regression.

TABLE 6 Performance comparison of models based on MAPE test statistics.

Inputs	SVR	RNN	LSTM	Best model
CAD/USD	1.8482	0.1709	0.2617	RNN
CNH/USD	3.038	0.0676	0.2019	RNN
CZK/USD	3.038	0.5549	0.3143	LSTM
HUF/USD	5.2005	0.3009	0.2645	LSTM
JPY/USD	1.8953	0.0776	0.0702	LSTM
MXN/USD	5.675	1.9669	1.9664	RNN/LSTM
NOK/USD	5.633	0.9866	0.7769	LSTM

Abbreviations: LSTM, long short-term memory; MAPE, mean absolute percentage error; RNN, recurrent neural network; SVR, support vector regression.

frequency. As well as employing robustness tests can help to identify the limitations and weaknesses of forecasting models and to determine whether they are capable of accurately predicting exchange rates under a variety of conditions.

In consideration of the largest interest, the comparison of model performances has been observed with the same exchange rate trend but with 30-min frequency interval. In comparison with LSTM with other benchmark models, LSTM tends to fit in with nonlinear trends more than any other learning model. This means the LSTM outperformed with respect to 30-min frequency interval time series and performs best in a larger context. This has been confirmed previously where in the context of large-scale financial market prediction tasks, LSTM extracts meaningful information in time series (Fischert, 2018). This study revealed that the prediction power of LSTM is best maybe because it can learn long-term dependencies with forget gates. The input and output gates in its architecture with a cell state in it regularize how much information has to pass on to the next level (Roy & Mukherjee, 2020).

T A B L E 7 Summary comparison of model fitness for multi-frequency intraday of selective exchange rates based on accuracy measures.

Currencies	Volatility	MSE			RMSE			MAE			MAPE		
		SVR	RNN	LSTM	SVR	RNN	LSTM	SVR	RNN	LSTM	SVR	RNN	LSTM
Frequency time interval of 1 h													
CAD/USD	Low	0.0012	1.1145	1.4111	0.0346	0.0033	0.0058	0.0245	0.0023	0.0036	1.8482	0.1709	0.2617
CNH/USD	Low	0.0619	3.6575	0.0002	0.2487	0.006	0.0167	0.2039	0.0046	0.0141	3.038	0.0676	0.2019
CZK/USD	High	0.0619	0.0448	0.0172	0.2487	0.2116	0.1312	0.2039	0.1358	0.0768	3.038	0.5549	0.3143
HUF/USD	High	348.858	1.969	1.2479	18.6777	1.4032	1.1171	14.8264	0.9482	0.8264	5.2005	0.3009	0.2645
JPY/USD	High	6.2519	0.0144	0.0135	2.5004	0.12	0.1162	2.0597	0.0822	0.0744	1.8953	0.0776	0.0702
MXN/USD	High	2.2985	0.5065	0.5065	1.5161	0.7117	0.7117	1.1722	0.46	0.4598	5.675	1.9669	1.9664
NOK/USD	Low	0.4257	0.0474	0.031	0.6524	0.2176	0.176	0.4982	0.1025	0.0808	5.633	0.9866	0.7769
Frequency time interval of 30 min													
CAD/USD	Low	0.0012	4.4995	5.7521	0.0349	0.0067	0.0024	0.0246	0.0042	0.0019	1.8576	0.3048	0.1382
CNH/USD	Low	0.0624	0.0002	0.0002	0.2497	0.013	0.0133	0.0267	0.0116	0.0127	3.0682	0.1667	0.1834
CZK/USD	High	1.0947	0.0319	0.0173	1.0463	0.1785	0.1314	0.2061	0.1025	0.0965	3.3356	0.4176	0.4025
HUF/USD	High	355.01	7.2786	1.2497	18.8416	0.0085	0.0011	14.9938	0.0083	0.0011	5.2532	0.1075	0.0135
JPY/USD	High	6.2982	0.0321	0.0092	2.5096	0.179	0.0957	2.0672	0.1492	0.0647	1.9042	0.1414	0.0605
MXN/USD	High	2.325	0.7558	0.6505	1.5248	0.8693	0.8065	1.1761	0.5525	0.5159	5.6839	2.357	2.2032
NOK/USD	Low	0.437	0.0622	0.0501	0.661	0.2495	0.2239	0.5038	0.1447	0.1144	5.6836	1.4191	1.1065

Note: Table 7 exhibits the range of currencies (for complete detail, see Table 1); their respective volatility, that is, Low = less volatile and High = more volatile; proposed supervised and unsupervised model range, that is, SVR, RNN, and LSTM; and accuracy indicators, that is, MSE, RMSE, MAE, and MAPE. The upper portion represents the statistics based on the frequency interval of 1-h intraday, and the lower portion represents the summary statistics based on 30-min frequency interval. Whereas, the values shown in bold indicate the lowest error values of test statistics.

Abbreviations: LSTM, long short-term memory; MAE, mean absolute error; MAPE, mean absolute percentage error; RMSE, root mean square error; RNN, recurrent neural network; SVR, support vector regression.

The graphical visualization for the analysis of 30 min time series frequency has been excluded to save space. However, the analysis of 1 h and 30-min frequency interval time series along with their comparison statistics for the best model has been provided in Table 5. Conclusively, this study suggests users use LSTM for financial market prediction tasks intraday because the deep learning-LSTM model has more predictability power in a larger scale of data series as shown in Table 7, further confirmed previously (Sun et al., 2020; Yan & Ouyang, 2018). LSTM model is superior for financial time series as exhibited in the study findings, as LSTM minimizes the error in the prediction of time series volatile behavior of similar nature.

4 | PRACTICAL IMPLICATIONS

The foreign exchange market is one of the largest trading markets to facilitate currency traders and is termed as a backbone of international traders. Because the economy directly or indirectly connects to exchange currency rates, the researchers, practitioners, and traders-related policies need to reflect these exchange rates using different predicting techniques. In the recent past due to the improvements in technology, there are advances in the field of artificial intelligence and ML. These ML algorithms are outperforming in the field of economics and finance as well, and all stakeholders of the currency market are getting benefits from the machine and deep learning techniques to forecast their future investments to diversify the risks and seek investment-related opportunities.

This study compared the most suggested models of machine and deep learning algorithms by using standard parameters such as batch size, epochs, neuron numbers, and activation functions for model tuning for predicting currency exchange rates of multi-frequency intervals. Based on empirical findings, this study suggests that researchers, academicians, practitioners, and investors use LSTM deep learning algorithm compared with other benchmark models. This model extracts meaningful information in time series with forget gate, despite of input and output gate in its architecture. Furthermore, there is a cell state in it that regularizes how much information has to pass on next level that produces the best prediction results for the observed time series with precision.

The use of forecasting techniques helps to eliminate market uncertainty (Erçen et al., 2022). To the traders and the investor, LSTM can provide them valuable insight into forex market trends, and informational patterns, and help them make more informed trading and investment decisions. By analyzing high-frequency data,

LSTM can potentially identify hidden patterns and relationships that may not be apparent to humans. Not limited to this, regulators and central banks can also use these techniques to identify hidden patterns and regulate the market fairly and transparently, so that no one can take the advantage of manipulation and insider trading. Additionally, these techniques can be used to stabilize the market and maintain price stability.

This study suggests all of these stakeholders of the currency market use LSTM practically for financial market prediction tasks for their investment decisions and investment-related policymaking in intraday because deep learning-LSTM model has more predictability power in the larger scale of data series. This has been confirmed through empirical evidence that the LSTM is best for time series as purposed earlier by Jung and Choi (2021) and Maneejuk and Srichaikul (2021). The results of the study help to improve decision-making by accurate exchange rate forecasts that can help businesses and investors make better decisions about when to buy or sell foreign currencies. This can help to mitigate risk and improve returns on investment. Furthermore, it will help policymakers to make better decisions about monetary and fiscal policies based on forecasted results using ML (Nikolopoulos et al., 2021; Syrowatka et al., 2021). This can further lead to improved economic performance, including increased economic growth and reduced inflation. Exchange rate volatility can lead to financial instability Bush and López Noria (2021), as sudden movements in exchange rates can lead to large losses for financial institutions. Accurate exchange rate forecasts can help to reduce this instability, by allowing institutions to better manage their exposure to currency risk.

5 | CONCLUSION

Almost every economy directly or indirectly has a link to the foreign exchange rate. Forecasting of foreign currency prices is a challenge for the stakeholders of forex market. Forecasting is a need of almost every industry due to the rapid improvements in technology recently. Whether a firm works for the customer demands, car industry, mobile phones, or the currency exchange rate, they need to forecast the future market. In the recent era, artificial intelligence in general and ML technology in specific is helping researchers, practitioners, and investors to predict currency rates. Machine and deep learning have captured all the phenomena, but the search for the best does not stop. This study has used Historical data 7 of currencies for 3 years, that is, January 1, 2018, to December 31, 2020. The time series data used in this study comprise

two intraday frequencies such as 1 h and 30 min. There are four accuracy indicators used for the evaluation criteria of the model performance. This study has used a standard tuning procedure for each currency analysis. The results can be improved if some other tuning process is used like an increase in batch size, neurons, layers, or activation function. For making results more improved, further studies can change the objectives of their research. There are different dynamics in every market, and different parameters are needed to handle such dynamics accordingly over time.

Initially, the MODWT analysis has been done to identify the real trend in each currency time series compared with the original time series trend. Three models have been selected in this study to predict each currency's intraday prices after reviewing previous studies done in a similar context. The most suggested machine and deep learning models that are selected are known as SVR, RNN, and LSTM. After the data preprocessing and environment selection, the predictability power of these models has been tested. The empirical results of tested models and the accuracy of these techniques than checked by four accuracy indicators such as MSE, MAE, RMSE and MAPE that have been extensively used previously. These results show that deep learning LSTM outperforms other benchmark models in predicting exercise. The study results are in line with the past literature, which has suggested that the prediction performance can be improved with deep learning networks as they can extract additional information (Abedin et al., 2021; Chong et al., 2017; Escudero et al., 2021; Ranjit et al., 2018; Sezer et al., 2020; Sun et al., 2020). Moreover, using deep learning techniques provides accuracy in prediction results and boosts performance too (Chatzis et al., 2018).

The results of this study can help traders, investors, and policymaker to improve their decision-making through accurate exchange rate forecasts. This will help to mitigate the risk and improve returns on investment. Furthermore, it will help policymakers to make better decisions about monetary and fiscal policies based on forecasted results using ML. This study has data limitation as the aforementioned data source only allows to retrieve a maximum of 3 years of data. For future work directions, this study suggests scholars use high-frequency data that include tick observations of the time series and addressed more volatile intraday exchange rates. Through the usage of advanced technology, high-frequency per-tick data can be processed more efficiently with the surety of timely and authentic results for currency traders. Furthermore, the usage of hybrid models for the prediction of exchange rates, with the inclusion of more volatile currencies and with updated tuning

parameters, would be of more interest to currency traders for strategy formulation and hedging purposes.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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